

Skills Summary

Distribution & Like-Term Errors

Use this reference while completing your worksheet or when studying.

Skill 1 — Distribute to every term

Rule: $a(b + c) = ab + ac$ — the outside factor multiplies *every* term inside the parentheses, never just the first one.

Example: Expand $4(2x + 3)$

Step 1. The 4 must reach both terms inside the parentheses, so multiply each term by 4 separately before adding.

Step 2. $4 \cdot 2x = 8x$ and $4 \cdot 3 = 12$.

$$\Rightarrow 4(2x + 3) = 8x + 12$$

Try it: Expand $3(4x + 5)$

$$12x + 15$$

(!) **Watch out:** the classic error is multiplying only the first term. If the outside factor is negative, every inside term also flips its sign.

Skill 2 — Combine only like terms

Rule: like terms have the *same variable part*. Add or subtract the coefficients; the variable part stays. The variable never disappears: $ax - bx = (a - b)x$.

Example: Simplify $7x - 2x + 4$

Step 1. Only the two x -terms are alike, so combine their coefficients and leave the constant term alone.

Step 2. $7x - 2x = 5x$; the $+4$ has no like partner, so it stays.

$$\Rightarrow 7x - 2x + 4 = 5x + 4$$

Try it: Simplify $9x - 5x + 2$

$$4x + 2$$

Skill 3 — Test equivalence with a number

Rule: equivalent expressions give the *same value for every* x . To expose a suspected error, substitute one number into both expressions: different values prove they are NOT equivalent.

Example: Is $2(x + 3)$ the same as $2x + 3$?

Step 1. A single test value is enough to disprove equivalence, so substitute an easy number into both expressions and compare.

Step 2. At $x = 4$: $2(4 + 3) = 14$, but $2 \cdot 4 + 3 = 11$. Different values — not equivalent.

Try it: Test $x = 3$ to decide whether $5(x + 1)$ and $5x + 5$ are equivalent.

check: both give **02** — equivalent

(!) Watch out: one matching value *suggests* equivalence but does not prove it; one *different* value proves non-equivalence for certain.